

|                                             |                                                                          |
|---------------------------------------------|--------------------------------------------------------------------------|
| Program: <b>Diploma in Renewable Energy</b> |                                                                          |
| Course Code: <b>4296</b>                    | Course Title: <b>Solar and Household Appliances Maintenance Workshop</b> |
| Semester: <b>4</b>                          | Credits: <b>1.5</b>                                                      |
| Course Category: <b>Program Core</b>        |                                                                          |
| Periods per week: <b>3 (L:0, T:0, P:3)</b>  | Periods per semester: <b>45</b>                                          |

### Course Objectives:

- To familiarize safety precautions practiced in the workplace.
- Understand the constructional details of solar appliances.
- To dismantle, assemble and repair household appliances.
- To assemble, install and operate solar devices.

### Course Prerequisites:

| Topic                                                                 | Course Code | Course Name                                     | Semester |
|-----------------------------------------------------------------------|-------------|-------------------------------------------------|----------|
| Charging methods and interconnections of batteries                    |             | Basic Electrical Technology & Renewable Sources | II       |
| Principle and working of solar Thermal and Solar photovoltaic devices |             | Solar Energy Technology                         | III      |

### Course Outcomes:

On completion of the course student will be able to:

| CO <sub>n</sub> | Description                                                                                     | Duration (Hours) | Cognitive Level |
|-----------------|-------------------------------------------------------------------------------------------------|------------------|-----------------|
| CO1             | Identify the parts, assemble and repair household appliances(Ceiling fan, Mixer, Electric iron) | 10               | Applying        |
| CO2             | Construct & carry out maintenance of Solar devices                                              | 11               | Applying        |
| CO3             | Carry out maintenance of batteries and inverters                                                | 9                | Applying        |
| CO4             | Carry out maintenance of Solar PV system                                                        | 9                | Applying        |
|                 | Lab Test                                                                                        | 6                |                 |

## CO – PO Mapping

| Course Outcomes | PO1 | PO2 | PO3 | PO4 | PO5 | PO6 | PO7 |
|-----------------|-----|-----|-----|-----|-----|-----|-----|
| CO1             |     | 3   |     | 3   |     |     |     |
| CO2             |     | 3   |     | 3   | 3   |     |     |
| CO3             |     | 3   |     | 3   |     |     |     |
| CO4             |     | 3   |     | 3   | 3   |     |     |

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

## Course Outline:

| Module Outcomes | Name of Experiment                                                           | Duration (Hours) | Cognitive Level |
|-----------------|------------------------------------------------------------------------------|------------------|-----------------|
| <b>CO1</b>      | <b>Identify the parts and maintenance of household electrical appliances</b> |                  |                 |
| M1.01           | Identify the faults and repair a given electric iron                         | 3                | Applying        |
| M1.02           | Dismantle and reassemble a given ceiling fan                                 | 3                | Applying        |
| M1.03           | Identify the parts,faults and repair a given mixer grinder                   | 3                | Applying        |
| <b>CO2</b>      | <b>Identify various solar appliances and conduct their maintenance</b>       |                  |                 |
| M2.01           | Assemble a solar cooker and carry out its maintenance                        | 3                | Applying        |
| M2.02           | Identify the parts,faults and repair a given solar water heater              | 3                | Applying        |
| M2.03           | Build the procedure of installing a solar water heater                       | 3                | Applying        |
| M 2.04          | Identify the parts,faults and repair a given solar pump                      | 3                | Applying        |
|                 | Lab Test-1                                                                   | 3                |                 |

|            |                                                                                        |   |               |
|------------|----------------------------------------------------------------------------------------|---|---------------|
| <b>CO3</b> | <b>Identify the procedures for maintenance of batteries and inverters</b>              |   |               |
| M3.01      | Carry out the maintenance of a given Lead acid battery                                 | 3 | Applying      |
| M3.02      | Identify the internal structure and maintenance of a Lithium Iron battery              | 3 | Applying      |
| M3.03      | Identify various parts , faults and repair a given inverter                            | 3 | Applying      |
| <b>CO4</b> | <b>Identify the maintenance of Solar Power Plant</b>                                   |   |               |
| M4.01      | Identify the parts of a standard solar charge controller                               | 3 | Understanding |
| M4.02      | Carry out maintenance of a Solar PV System                                             | 3 | Applying      |
| M4.03      | Identify various parts and carry out the maintenance of Solar Power Plant (up to 5kWp) | 3 | Applying      |
|            | Lab test -2                                                                            | 3 |               |

### Open ended Experiments:

Prepare an estimate for Solar PV Street Light Installation.  
Prepare troubleshooting chart for various electrical equipments

### Text / Reference

| <b>T/R</b> | <b>Book Title/Author</b>                                                                                                              |
|------------|---------------------------------------------------------------------------------------------------------------------------------------|
| T1         | From Sunlight to Electricity,A Practical Handbook on Solar Photovoltaic Applications-Third Edition-Suneel Deambi                      |
| T2         | Solar Photovoltaic Technology and Systems: A Manual for Technicians, Trainers and Engineers Chetan Singh Solanki-PHI learning Pvt Ltd |
| R1         | Solar Energy Utilisation, G. D. Rai, Khanna Publishers                                                                                |

### Online Resources

| Sl.No | Website Link                                                                                                                      |
|-------|-----------------------------------------------------------------------------------------------------------------------------------|
| 1     | <a href="#">Home Appliance Repair Textbooks Collection : Lanceshter : Free Download, Borrow, and Streaming : Internet Archive</a> |
| 2     | <a href="http://www.jnkvv.org/PDF/30042020142936304201343.pdf">http://www.jnkvv.org/PDF/30042020142936304201343.pdf</a>           |

### Assessment Questions:

1. Illustrate the parts of the given appliance.
2. Identify the fault on the given appliance and repair it for proper working
3. Assemble a solar cooker
4. Install a stand alone PV system for supplying an AC load
5. Install a stand alone PV system for supplying an DC load.